UNIT

FOUR

Straight lines and planes in space

Lesson 1 Equation of a straight line in space.

Lesson 2 Equation of a plane in space.

Lesson 3 Relative positions of two planes in space.



Unit 4

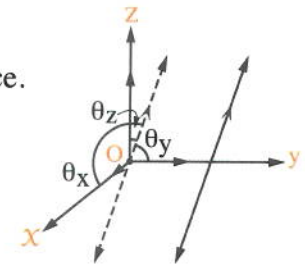
Lesson One

Equation of a straight line in space

Direction vector of a straight line in space

- Its known that the directed angles of a straight line in the space passing through the origin point are the measures of the angles that the straight line makes with the positive directions of the coordinate axes.
- The directed angles of a straight line does not pass through the origin point are the directed angles of a straight line passing through the origin and parallel to this line.

In the opposite figure :



- θ_x , θ_y and θ_z are the directed angles of the straight line in the space.
- $\cos\theta_x$, $\cos\theta_y$ and $\cos\theta_z$ are the directed cosines of the straight line and they are usually denoted by ℓ , m , n where $\ell = \cos\theta_x$, $m = \cos\theta_y$ and $n = \cos\theta_z$
- The vector $\vec{u} = \ell\,\hat{i} + m\,\hat{j} + n\,\hat{k}$ is the unit vector in the direction of the straight line and hence $\ell^2 + m^2 + n^2 = 1$
- The direction vector of the straight line is any vector parallel to the unit vector in the direction of the straight line and is denoted by $\vec{d}$

$\therefore \vec{d} = k\,\vec{u} = k\,(\ell\,\hat{i} + m\,\hat{j} + n\,\hat{k})$ "where $k \in \mathbb{R}^*$"

$\therefore \vec{d}$ can by written as $\vec{d} = (a , b , c)$ where $a = k\,\ell$, $b = k\,m$, $c = k\,n$, $\because$ a , b and c are proportional with ℓ , m , n , then a , b and c are called dirction ratios of the straight line.

For example : If the directed angles of a straight line in the space are :

60° , 135° and 60° , then :

- $\left(\frac{1}{2} , -\frac{1}{\sqrt{2}} , \frac{1}{2}\right)$ are the directed cosines of the straight line.

- $\vec{u} = \left(\frac{1}{2}, -\frac{1}{\sqrt{2}}, \frac{1}{2}\right)$ is the unit vector in the direction of the straight line.
- $\vec{d} = k\left(\frac{1}{2}, -\frac{1}{\sqrt{2}}, \frac{1}{2}\right)$ is direction vector of the straight line where $k \in \mathbb{R}^*$

$\therefore$ For each $k \in \mathbb{R}^*$ there is a directed vector of the straight line.

i.e. The straight line has infinite number of parallel direction vectors each of them parallel to this line.

- $\frac{1}{2}k$, $-\frac{k}{\sqrt{2}}$, $\frac{1}{2}k$ are direction ratios of the straight line.

Remarks

1. If A and B are two points on the straight line , then :
- Direction vector of the straight line = $\overrightarrow{AB} = \vec{B} - \vec{A}$
- $k(\overrightarrow{AB})$ where $k \in \mathbb{R}^*$ is also direction vector of the same straight line.

For example : Direction vector of the straight line passing through A $(3, -2, 4)$, B $(-1, 1, -2)$ is $\overrightarrow{AB} = (-1, 1, -2) - (3, -2, 4) = (-4, 3, -6)$

Notice : $k(-4, 3, -6)$, $k \in \mathbb{R}^*$ is also direction vector of the same straight line.

2. If a straight line passing through the origin and the point A (x_1, y_1, z_1) , then : $\vec{A} = (x_1, y_1, z_1)$ is direction vector of the straight line.

3. $(a, 0, 0)$ is direction vector of the x-axis or (any straight line parallel to it) , $(0, b, 0)$ is direction vector of the y-axis or (any straight line parallel to it) and $(0, 0, c)$ is direction vector of the z-axis or (any straight line parallel to it).

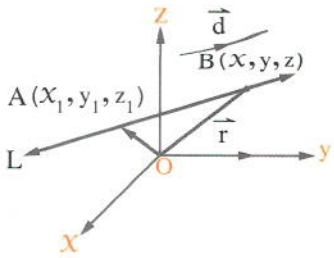
4. The straight line whose direction vector $\vec{d} = (a, b, 0)$ lies in a plane parallel to the plane xy , similarly the straight line whose direction vector $\vec{d} = (a, 0, c)$ lies in a plane parallel to the plane xz and the straight line whose direction vector $\vec{d} = (0, b, c)$ lies in a plane parallel to the plane yz

5. If $\vec{d}$ is a direction vector of the straight line , then $-\vec{d}$ is a direction vector of the same straight line and so directed angles $(\theta_x, \theta_y, \theta_z)$ and $(\pi - \theta_x, \pi - \theta_y, \pi - \theta_z)$ are directed angles of the same straight line.

6. Notice the difference between directed cosines of a straight line and its direction vector :
- ℓ, m, n are directed cosines of the straight line where (ℓ, m, n) is unit vector in the direction of the straight line and $\ell^2 + m^2 + n^2 = 1$
- a, b, c are the direction ratios of the straight line where (a, b, c) is direction vector of the straight line , $(a, b, c) = k(\ell, m, n)$, $k \neq 0$
- $(\ell, m, n) = \frac{(a, b, c)}{\sqrt{a^2 + b^2 + c^2}}$

Different forms of equation of the straight line in space

If L is a straight line in space where A (x_1 , y_1 , z_1) is a given point on it and $\vec{d} = (a , b , c)$ is a direction vector of it, then the position vector of any other point (B) on it is $\vec{r} = (x , y , z)$ is determine from the relation



$\vec{r} = \overrightarrow{OA} + \overrightarrow{AB}$ "from Δ OAB"

, $\because \vec{d} \,//\, \overrightarrow{AB}$ $\quad\quad$ $\therefore \overrightarrow{AB} = t\,\vec{d}$ where $t \in \mathbb{R}^*$

$\therefore$ $\vec{r} = \vec{A} + t\,\vec{d}$ "vector form of the equation of the straight line"

- $\vec{r}$: Position vector of any point on the straight line
- $\vec{A}$: Position vector of a given point on the straight line
- t : Real number
- $\vec{d}$: Direction vector of the straight line

$\therefore (x , y , z) = (x_1 , y_1 , z_1) + t\,(a , b , c)$

Where t is not a certain real number but it takes different real values therefore t is called a parameter and for each real value of this parameter we get a point on the straight line.

Notice that
In case t = 0 $\vec{r}$ represents the position vector of $\vec{A}$ itself.

$\therefore$ $x = x_1 + t\,a \;,\; y = y_1 + t\,b \;,\; z = z_1 + t\,c$ "Parametric equations of the straight line"

$\therefore\ t = \dfrac{x - x_1}{a}$, $t = \dfrac{y - y_1}{b}$, $t = \dfrac{z - z_1}{c}$

i.e. $\dfrac{x - x_1}{a} = \dfrac{y - y_1}{b} = \dfrac{z - z_1}{c}$ "Cartesian equation of the straight line"

Remarks

By substituting in the parametric equation to deduce, the cartesian equation we find :

- In case of a = 0 the cartesian equation is : $\dfrac{y - y_1}{b} = \dfrac{z - z_1}{c}$, $x = x_1$
- In case of b = 0 , the cartesian equation is : $\dfrac{x - x_1}{a} = \dfrac{z - z_1}{c}$, $y = y_1$
- In case of c = 0 , the cartesian equation is : $\dfrac{x - x_1}{a} = \dfrac{y - y_1}{b}$, $z = z_1$
- Where direction vector of the straight line a , b , c are proportional with directed cosine ℓ , m , n , then the cartesian form of the straight line is $\dfrac{x - x_1}{\ell} = \dfrac{y - y_1}{m} = \dfrac{z - z_1}{n}$

- Equation of $\mathcal{X}$-axis is : $y = 0$, $z = 0$
  , equation of y-axis is : $\mathcal{X} = 0$, $z = 0$
  , equation of z-axis is : $\mathcal{X} = 0$, $y = 0$
- Equation of any straight line parallel to $\mathcal{X}$-axis and passes through the point $(\mathcal{X}_1 , y_1 , z_1)$ is : $y = y_1$, $z = z_1$
- Equation of any straight line parallel to y-axis and passes through the point $(\mathcal{X}_1 , y_1 , z_1)$ is : $\mathcal{X} = \mathcal{X}_1$, $z = z_1$
- Equation of any straight line parallel to z-axis and passes through the point $(\mathcal{X}_1 , y_1 , z_1)$ is : $\mathcal{X} = \mathcal{X}_1$, $y = y_1$
- Different forms of the equation of the straight line passing through the origin and the vector (a , b , c) is direction vector are :
  * $\vec{r} = t\,(a , b , c)$ "vector form"
  * $\mathcal{X} = t\,a$, $y = t\,b$, $z = t\,c$ "parametric equations"
  * $\frac{\mathcal{X}}{a} = \frac{y}{b} = \frac{z}{c}$ "cartesian form"
- Parametric equations of the straight line are used to find the intersection point / points of this straight line with a plane.

Example 1

Find the different forms of the equation of the straight line:

1. Passing through the point $(2 , 3 , -1)$ and the vector $(-1 , 2 , 3)$ is a direction vector to it

2. Passing through the two points $(-5 , 2 , 3)$ and $(7 , 0 , -1)$

3. Passing through the point $(-3 , 4 , 5)$ and its directed angles $(30° , 90° , 60°)$

4. Passing through the point $(3 , 4 , 1)$ parallel to $\mathcal{X}$-axis.

5. Passing through the point $(-2 , 1 , 3)$ and makes equal angles with the positive direction with the coordinate axis.

Solution

1.
- Vector form : $\vec{r} = (2 , 3 , -1) + t\,(-1 , 2 , 3)$
- Parametric equations :

  $\mathcal{X} = 2 - t$, $y = 3 + 2t$, $z = -1 + 3t$
- Cartesian form : $\frac{\mathcal{X} - 2}{-1} = \frac{y - 3}{2} = \frac{z + 1}{3}$

2. Direction vector of the straight line $= (7\ ,0\ ,-1) - (-5\ ,2\ ,3) = (12\ ,-2\ ,-4)$

- Vector form : $\vec{r} = (7\ ,0\ ,-1) + t\,(12\ ,-2\ ,-4)$
- Parametric equations : $x = 7 + 12t\ ,\ y = -2t\ ,\ z = -1 - 4t$
- Cartesion form : $\frac{x-7}{12} = \frac{y}{-2} = \frac{z+1}{-4}$

3. Unit vector in the direction of the straight line $= (\cos 30°\ ,\cos 90°\ ,\cos 60°)$

$$= \left(\frac{\sqrt{3}}{2}\ ,0\ ,\frac{1}{2}\right)$$

$\therefore \left(\sqrt{3}\ ,0,1\right)$ is a direction vector for the same straight line

- Vector form : $\vec{r} = (-3\ ,4\ ,5) + t\left(\sqrt{3}\ ,0,1\right)$
- Parametric equations : $x = -3 + \sqrt{3}\,t\ ,\ y = 4\ ,\ z = 5 + t$
- Cartesian form : $y = 4\ ,\ \frac{x+3}{\sqrt{3}} = \frac{z-5}{1}$

i.e. $y = 4\ ,\ x + 3 = \sqrt{3}\,(z-5)$

4. Direction vector of the straight line $= (1\ ,0\ ,0)$

- Vector form : $\vec{r} = (3\ ,4\ ,1) + t\,(1\ ,0\ ,0)$
- Parametric equations : $x = 3 + t\ ,\ y = 4\ ,\ z = 1$
- Cartesian form : $y = 4\ ,\ z = 1$

5. $\because$ The straight line makes equal angles with the positive direction of the coordinate axes

$\therefore \cos\theta_x = \cos\theta_y = \cos\theta_z$

, $\because \cos^2\theta_x + \cos^2\theta_y + \cos^2\theta_z = 1 \quad \therefore \cos^2\theta_x = \cos^2\theta_y = \cos^2\theta_z = \frac{1}{3}$

$\therefore \cos\theta_x = \cos\theta_y = \cos\theta_z = \pm\frac{1}{\sqrt{3}}$

$\therefore \left(\frac{1}{\sqrt{3}}\ ,\frac{1}{\sqrt{3}}\ ,\frac{1}{\sqrt{3}}\right)$ is a direction vector of the straight line

$\therefore (1\ ,1\ ,1)$ is also direction vector of the straight line

- Direction form of the equation : $\vec{r} = (-2\ ,1\ ,3) + t\,(1\ ,1\ ,1)$
- Parametric equations : $x = -2 + t\ ,\ y = 1 + t\ ,\ z = 3 + t$
- Cartesian form : $x + 2 = y - 1 = z - 3$

Example 2

Find the different forms of the equation of the straight line and the coordinate of a point on it in each of the following cases :

1. It passes through the origin and 2 , – 1 and 7 are direction ratios for it

2. Its equation in vector form is : $\vec{r} = (-1 , 2 , 5) + t\,(-1 , 4 , 3)$

3. Its cartesian equation is : $3x = 4y = \frac{-z+1}{5}$

4. Its cartesian equation is : $y = 5$, $\frac{x-4}{3} = \frac{2z+1}{6}$

Solution

1.
- Vector form : $\vec{r} = t\,(2 , -1 , 7)$
- Parametric equations : $x = 2t$, $y = -t$, $z = 7t$
- Cartesian form : $\frac{x}{2} = \frac{y}{-1} = \frac{z}{7}$
- Putting $x = 2$ $\therefore y = -1$, $z = 7$

$\therefore (2 , -1 , 7)$ is a point on the straight line.

2.
- Parametric equations : $x = -1 - t$, $y = 2 + 4t$, $z = 5 + 3t$
- Cartesian equation : $\frac{x+1}{-1} = \frac{y-2}{4} = \frac{z-5}{3}$
- Putting $x = -1$ $\therefore y = 2$, $z = 5$

$\therefore (-1 , 2 , 5)$ is a point on the straight line.

3.
- Putting $3x = 4y = \frac{-z+1}{5} = t$, then we deduce that
- $x = \frac{1}{3}t$, $y = \frac{1}{4}t$, $z = 1 - 5t$ are parametric equations of the straight line
- $\vec{r} = (0 , 0 , 1) + t\left(\frac{1}{3} , \frac{1}{4} , -5\right)$ is the vector form of the equation of the straight line
- Putting $t = 24$ $\therefore x = 8$, $\therefore y = 6$, $z = -119$

$\therefore (8 , 6 , -119)$ is a point on the straight line.

4.
- Putting $\frac{x-4}{3} = \frac{2z+1}{6} = t$
- $x = 4 + 3t$, $y = 5$, $z = -\frac{1}{2} + 3t$ are the parametric equation of the straight line
- $\vec{r} = \left(4 , 5 , -\frac{1}{2}\right) + t\,(3 , 0 , 3)$ is the vector form of the equation of the straight line
- Putting $t = 1$ $\therefore x = 7$, $\therefore z = 2\frac{1}{2}$, $y = 5$

$\therefore \left(7 , 5 , 2\frac{1}{2}\right)$ is a point on the straight line.

Example 3

Find the equation of the straight line :

1. Passing through the point A $(2 , 3 , -5)$ parallel to the straight line passing through the two points B $(-1 , 3 , 2)$ and C $(2 , -1 , 3)$, then prove that it passes through the point $(17 , -17 , 0)$

2. Passing through the point $(-2 , 3 , 4)$ parallel to the straight line $\frac{x-2}{3} = \frac{2y+1}{6} = \frac{z-4}{2}$

3. Passing through the intersection point of the two straight lines :

$L_1 : \vec{r_1} = (1 , -1 , 2) + t_1 , (3 , -2 , 5)$ and $L_2 : \vec{r_2} = (8 , -2 , 14) + t_2 (1 , 3 , 2)$

and the point $(2 , -3 , 1)$

Solution

1. ∵ The required straight line parallel to $\overleftrightarrow{BC}$

∴ $\overrightarrow{BC}$ is a direction vector of it

, $\overrightarrow{BC} = (2 , -1 , 3) - (-1 , 3 , 2) = (3 , -4 , 1)$

- Vector form of the equation of the straight line : $\vec{r} = (2 , 3 , -5) + t (3 , -4 , 1)$
- Parametric equations : $x = 2 + 3t$, $y = 3 - 4t$, $z = -5 + t$
- Cartesian form : $\frac{x-2}{3} = \frac{y-3}{-4} = z + 5$

, ∵ $\frac{17-2}{3} = 5$, $\frac{-17-3}{-4} = 5$, $0 + 5 = 5$

∴ The straight line passes through the point $(17 , -17 , 0)$

2. ∵ Equation of the given straight line is :

$\frac{x-2}{3} = \frac{2y+1}{6} = \frac{z-4}{2}$

i.e. $\frac{x-2}{3} = \frac{y+\frac{1}{2}}{3} = \frac{z-4}{2}$

∴ $(3 , 3 , 2)$ is a direction vector of the given line

, ∵ the required line is parallel to the given one

∴ $(3 , 3 , 2)$ is a direction vector of the required line.

- Vector form of the equation of the straight line : $\vec{r} = (-2 , 3 , 4) + t (3 , 3 , 2)$
- Parametric equations : $x = -2 + 3t$, $y = 3 + 3t$, $z = 4 + 2t$
- Cartesian form : $\frac{x+2}{3} = \frac{y-3}{3} = \frac{z-4}{2}$

Notice that

Its possible to put the cartesian equation in its standard form : $\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c}$

"**i.e.**" Making coefficients of x , y and z equal to one" to get (a , b , c) a direction vector of the straight line and (x_1 , y_1 , z_1) point on it

3. At the intersection point of the two straight lines we get $\overrightarrow{r_1} = \overrightarrow{r_2}$

$\therefore (1 , -1 , 2) + t_1 (3 , -2 , 5) = (8 , -2 , 14) + t_2 (1 , 3 , 2)$

$\therefore 1 + 3t_1 = 8 + t_2$ $\qquad \therefore 3t_1 - t_2 = 7$ (1)

$, -1 - 2t_1 = -2 + 3t_2$ $\qquad \therefore 2t_1 + 3t_2 = 1$ (2)

$, 2 + 5t_1 = 14 + 2t_2$ $\qquad \therefore 5t_1 - 2t_2 = 12$ (3)

, solving (1) , (2) : $\therefore t_1 = 2 , t_2 = -1$

, then we find the two solutions are satisfiying the equation (3) also

, substituting the value of t_1 in the equaiton of L_1

$\therefore$ The intersection point is : $(x , y , z) = (1 , -1 , 2) + 2 (3 , -2 , 5) = (7 , -5 , 12)$

$\therefore$ The vector $(2 , -3 , 1) - (7 , -5 , 12) = (-5 , 2 , -11)$ is direction vector of the required straight line

- vector form : $\overrightarrow{r} = (2 , -3 , 1) + t (-5 , 2 , -11)$
- parametric equations : $x = 2 - 5t$, $y = -3 + 2t$, $z = 1 - 11t$
- cartesian form : $\frac{x-2}{-5} = \frac{y+3}{2} = \frac{z-1}{-11}$

The angle between two straight lines in space

The angle measure between two straight lines in space is the measure of the smaller angle between them.

If L_1 and L_2 are two straight lines in space , their direction vectors are :

$\overrightarrow{d_1} = (a_1 , b_1 , c_1)$ and $\overrightarrow{d_2} = (a_2 , b_2 , c_2)$

, then the cosine of the smaller angle between the two straight lines L_1 and L_2 (θ) is given by the relation : $\cos\theta = \frac{|\overrightarrow{d_1} . \overrightarrow{d_2}|}{\|\overrightarrow{d_1}\| \, \|\overrightarrow{d_2}\|}$ and if (ℓ_1 , m_1 , n_1) and (ℓ_2 , m_2 , n_2) are the directed cosines of the two straight lines , then $\cos\theta = |\ell_1 \ell_2 + m_1 m_2 + n_1 n_2|$

Example 4

Find measure of the angle between the two straight lines :

1. L_1 : passing through the two points $(5 , 3 , 5)$ and $(-1 , 7 , 3)$

, L_2 : passing through $(2 , 5 , 4)$, $(-2 , 3 , 1)$

2. $L_1 : \overrightarrow{r} = (2 , -3 , 4) + t (1 , 2 , 5)$, $L_2 : \frac{x-3}{2} = \frac{y+4}{1} = \frac{z+6}{4}$

3. If their directed cosines are : $\left(\frac{1}{2} , -\frac{1}{\sqrt{2}} , \frac{1}{2}\right)$ and $\left(\frac{\sqrt{3}}{2} , \frac{1}{2\sqrt{2}} , \frac{-1}{2\sqrt{2}}\right)$

Solution

1. $\overrightarrow{d_1} = (-1\ ,7\ ,3) - (5\ ,3\ ,5) = (-6\ ,4\ ,-2)$

$, \overrightarrow{d_2} = (-2\ ,3\ ,1) - (2\ ,5\ ,4) = (-4\ ,-2\ ,-3)$

$\because \cos\theta = \dfrac{|\overrightarrow{d_1} \cdot \overrightarrow{d_2}|}{\|\overrightarrow{d_1}\|\ \|\overrightarrow{d_2}\|}$

$\therefore \cos\theta = \dfrac{|(-6\ ,4\ ,-2) \cdot (-4\ ,-2\ ,-3)|}{\sqrt{(-6)^2+(4)^2+(-2)^2}\ \sqrt{(-4)^2+(-2)^2+(-3)^2}} = \dfrac{|24-8+6|}{2\sqrt{14} \times \sqrt{29}} = \dfrac{11}{\sqrt{406}}$

$\therefore \theta \simeq 56^\circ\ 54\grave{}\ 45\grave{}\grave{}$

2. $\overrightarrow{d_1} = (1\ ,2\ ,5)\ ,\ \overrightarrow{d_2} = (2\ ,1\ ,4)$

$, \because \cos\theta = \dfrac{|\overrightarrow{d_1} \cdot \overrightarrow{d_2}|}{\|\overrightarrow{d_1}\|\ \|\overrightarrow{d_2}\|}$

$\therefore \cos\theta = \dfrac{|(1\ ,2\ ,5) \cdot (2\ ,1\ ,4)|}{\sqrt{1+4+25}\ \sqrt{4+1+16}} = \dfrac{|2+2+20|}{\sqrt{30}\ \sqrt{21}} = \dfrac{8}{\sqrt{70}}$

$\therefore \theta \simeq 17^\circ\ 1\grave{}\ 26\grave{}\grave{}$

3. $(\ell_1\ , m_1\ , n_1) = \left(\frac{1}{2}\ , -\frac{1}{\sqrt{2}}\ , \frac{1}{2}\right)\ , \left(\ell_2\ , m_2\ , n_2\right) = \left(\frac{\sqrt{3}}{2}\ , \frac{1}{2\sqrt{2}}\ , \frac{-1}{2\sqrt{2}}\right)$

$, \because \cos\theta = |\ell_1\ \ell_2 + m_1\ m_2 + n_1\ n_2|$

$\therefore \cos\theta = \left|\left(\frac{1}{2}\right)\left(\frac{\sqrt{3}}{2}\right) + \left(-\frac{1}{\sqrt{2}}\right)\left(\frac{1}{2\sqrt{2}}\right) + \left(\frac{1}{2}\right)\left(\frac{-1}{2\sqrt{2}}\right)\right| = \left|\frac{\sqrt{6}-\sqrt{2}-1}{4\sqrt{2}}\right|$

$\therefore \theta \simeq 89^\circ\ 38\grave{}\ 34\grave{}\grave{}$

Two parallel lines in space

If $\overrightarrow{d_1} = (a_1\ , b_1\ , c_1)$ and $\overrightarrow{d_2} = (a_2\ , b_2\ , c_2)$

are two direction vectors of the two straight lines L_1 and L_2 respectively , then :

$L_1 \,//\, L_2$ if and only if $\overrightarrow{d_1} // \overrightarrow{d_2}$

, the previous condition can be satisfied by several forms :

1. $\overrightarrow{d_1} = t\,\overrightarrow{d_2}\ , t \in \mathbb{R}^*$ **2.** $\dfrac{a_1}{a_2} = \dfrac{b_1}{b_2} = \dfrac{c_1}{c_2}$ **3.** $\overrightarrow{d_1} \times \overrightarrow{d_2} = \overrightarrow{O}$

Remarks

1. If the two straight lines are parallel and there is a point on one of them satisfying the equation of the other then the two straight lines are coincident.

2. If the two straight lines don't satisfy one of the previous conditions of parallel , then the two straight lines either intersecting or skew.

Example 5

If the two straight lines :

$L_1 : \vec{r} = (2 , 3 , -4) + t\,(2 , 3 , a)$, $L_2 : \frac{x-5}{b} = \frac{y+4}{6} = \frac{z-4}{2}$ are parallel find a , b

Solution

$(2 , 3 , a)$ is direction vector of the straight line L_1

, $(b , 6 , 2)$ is direction vector of the straight line L_2

, $\because L_1 // L_2$ $\quad \therefore \frac{2}{b} = \frac{3}{6} = \frac{a}{2}$ $\quad \therefore b = 4 , a = 1$

Perpendicular two lines in space

If $\vec{d_1} = (a_1 , b_1 , c_1)$ and $\vec{d_2} = (a_2 , b_2 , c_2)$

are two direction vectors of the two straight lines L_1 and L_2 respectively , then

$L_1 \perp L_2$ if and only if $\vec{d_1} . \vec{d_2} = 0$

i.e. $a_1 a_2 + b_1 b_2 + c_1 c_2 =$ zero and it is possible to use the directed cosines as a direction vector of each straight line

Remarks

1. The two parallel straight lines are contained in the same plane.
2. The two intersecting straight lines are contained in the same plane.
3. The two perpendicular straight lines : either are intersecting orthogonally and hence contained in the same plane or are skew and hence they are not contained in one plane.

Example 6

Prove that the two straight lines : $L_1 : \vec{r} = (1 , 2 , 4) + t_1\,(4 , -2 , 2)$,

$L_2 : x = 1 - 6\,t_2$, $y = 1 + 21\,t_2$, $z = 1 + 33\,t_2$ are perpendicular and skew.

Solution

$\vec{d_1}$ (direction vector of the straight line L_1) $= (4 , -2 , 2)$

, $\vec{d_2}$ (direction vector of the straight line L_2) $= (-6 , 21 , 33)$

$\therefore \vec{d_1} . \vec{d_2} = (4 , -2 , 2) \cdot (-6 , 21 , 33)$

$= 4 \times -6 + -2 \times 21 + 2 \times 33 =$ zero

$\therefore$ The two lines are perpendicular

• Finding the intersection point of the two lines

let C (x_1 , y_1 , z_1) is the point of intersection

$\therefore C \in L_1$ $\quad\quad$ $\therefore x = 1 + 4 t_1$

, $y = 2 - 2 t_1$ $\quad\quad$, $z = 4 + 2 t_1$

, $C \in L_2$ $\quad\quad$ $\therefore x = 1 - 6 t_2$

, $y = 1 + 21 t_2$ $\quad\quad$, $z = 1 + 33 t_2$

$\therefore 1 + 4 t_1 = 1 - 6 t_2$ and hence $2 t_1 + 3 t_2 = 0$ (1)

, $2 - 2 t_1 = 1 + 21 t_2$ and hence $2 t_1 + 21 t_2 = 1$ (2)

, $4 + 2 t_1 = 1 + 33 t_2$ and hence $2 t_1 - 33 t_2 = -3$ (3)

Solving (1) , (2) : $\therefore t_1 = -\frac{1}{12}$, $t_2 = \frac{1}{18}$

, then substituting in (3) :

$\therefore 2 \times -\frac{1}{12} - 33 \times \frac{1}{18} = -2 \neq -3$

$\therefore$ There are no t_1 and t_2 satisfying the equations (1) , (2) , (3) simultaneously

$\therefore$ The two lines are not intersecting

, $\because$ the two lines are perpendicular

$\therefore$ The two lines are skew and perpendicular

Notice that

In parametric equations of the straight line whenever the coefficients of x , y , z equal one , then the coefficients of t are the components of the direction vector.

Remarks

In the not parallel two straight lines

$\vec{r_1} = (x_1 , y_1 , z_1) + t_1 (a_1 , b_1 , c_1)$

and $\vec{r_2} = (x_2 , y_2 , z_2) + t_2 (a_2 , b_2 , c_2)$

1. If there are values of t_1 and t_2 satisfy that $\vec{r_1} = \vec{r_2}$, then the two lines are intersecting.
2. If there are not values of t_1 and t_2 satisfy that $\vec{r_1} = \vec{r_2}$, then the two lines are skew.

Another solution :

To prove that the two straight lines are skew

$L_1 : \vec{r_1} = (1 , 2 , 4) + t_1 (4 , -2 , 2)$

$L_2 : \vec{r_2} = (1 , 1 , 1) + t_2 (-6 , 21 , 33)$

$\overrightarrow{D_1 D_2} = (1 - 1 , 1 - 2 , 1 - 4) = (0 , -1 , -3)$

, $\because \overrightarrow{D_1 D_2} . (\vec{d_1} \times \vec{d_2}) = \begin{vmatrix} 0 & -1 & -3 \\ 4 & -2 & 2 \\ -6 & 21 & 33 \end{vmatrix}$

$= (4 \times 33 + 6 \times 2) - 3 (4 \times 21 - 6 \times 2) = -72 \neq$ zero

$\therefore$ The two straight lines are skew

Remark

The two straight lines

$L_1 : \vec{r_1} = \overrightarrow{D_1} + t_1 \vec{d_1}$

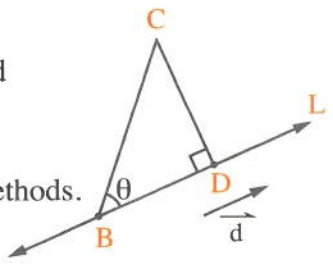
, $L_2 : \vec{r_2} = \overrightarrow{D_2} + t_2 \vec{d_2}$

are skew if

$\overrightarrow{D_1 D_2} . (\vec{d_1} \times \vec{d_2}) \neq$ zero

The length of the perpendicular drawn from a point to a straight line in space

Let L be a straight line in space , and its equation $\vec{r} = \vec{B} + t\,\vec{d}$ passes through the point B and its direction vector $\vec{d}$, then to find the distance between point C in the space and the straight line L (say CD) where $\overline{CD} \perp L$, $D \in L$, follow one of the following methods.



- **The first method**

1. Find $BC = \| \overrightarrow{BC} \|$

2. Find BD = The length of projection of $\overrightarrow{BC}$ in direction of $\vec{d} = \dfrac{|\overrightarrow{BC} \cdot \vec{d}|}{\|\vec{d}\|}$

3. Find the perpendicular length $(CD) = \sqrt{(BC)^2 - (BD)^2}$ by using Pythagoras.

- **The second method**

1. Find $BC = \| \overrightarrow{BC} \|$

2. Find the measure of angle θ from the formula $\cos\theta = \dfrac{|\overrightarrow{BC} \cdot \vec{d}|}{\|\overrightarrow{BC}\| \|\vec{d}\|}$

3. Find length of the perpendicular (CD) = BC sin θ

- **The third method**

1. $\because D \in L$

$\therefore$ D satisfies the equation of the straight line so you can write D in term of t

2. $\because \overrightarrow{CD} \perp L$

$\therefore \overrightarrow{CD} \cdot \vec{d}$ = zero and so you can find the value of t and so you can calculate the point D

3. Find length of the perpendicular $(CD) = \| \overrightarrow{CD} \|$

- **The fourth method**

$\because \| \overrightarrow{BC} \times \vec{d} \| = \| \vec{d} \| \| \overrightarrow{BC} \| \sin\theta$

, $\because$ the perpendicular length $= \| \overrightarrow{BC} \| \sin\theta$

$\therefore$ The perpendicular length (CD) $= \dfrac{\| \overrightarrow{BC} \times \vec{d} \|}{\|\vec{d}\|}$ and the following example shows that.

Remark

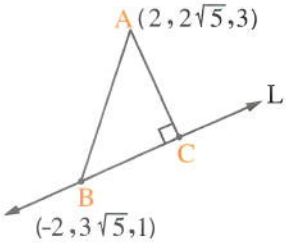
It is possible to use $\vec{d}$ "the direction vector" instead of $\overrightarrow{BA}$

Example 7

Find the distance between the point A $(2 , 2\sqrt{5} , 3)$ and the straight line L whose equation in the vector form is : $\vec{r} = (-2 , 3\sqrt{5} , 1) + t(8 , -\sqrt{5} , 2)$

Solution

$\because$ B $(-2 , 3\sqrt{5} , 1)$ is a point on the straight line L



$\therefore \overrightarrow{BA} = (2 , 2\sqrt{5} , 3) - (-2 , 3\sqrt{5} , 1) = (4 , -\sqrt{5} , 2)$

$\therefore$ BA $= \| \overrightarrow{BA} \| = \sqrt{4^2 + (-\sqrt{5})^2 + 2^2} = 5$ length unit

, BC = length of projection of $\overrightarrow{BA}$ on the straight line L

, $\because \vec{d} = (8 , -\sqrt{5} , 2)$ is direction vector of the straight line L

$$\therefore BC = \frac{|\overrightarrow{BA} . \vec{d}|}{\| \vec{d} \|} = \frac{|(4 , -\sqrt{5} , 2) . (8 , -\sqrt{5} , 2)|}{\sqrt{8^2 + (-\sqrt{5})^2 + 2^2}} = \frac{|32 + 5 + 4|}{\sqrt{64 + 5 + 4}} = \frac{41}{\sqrt{73}} \text{ length unit.}$$

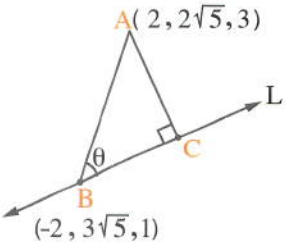
$$\therefore \text{Distance between the point A and the straight line L} = AC = \sqrt{5^2 - \left(\frac{41}{\sqrt{73}}\right)^2} = \frac{12}{\sqrt{73}}$$

$\simeq 1.4$ length unit.

Another solution :

$\because$ Direction vector of the straight line L is $\vec{d_1} = (8 , -\sqrt{5} , 2)$

, $\because$ direction vector of $\overleftrightarrow{AB}$ is $\vec{d_2} = (4 , -\sqrt{5} , 2)$



$$\therefore \cos\theta = \frac{|(8 , -\sqrt{5} , 2) . (4 , -\sqrt{5} , 2)|}{\sqrt{64 + 5 + 4}\ \sqrt{16 + 5 + 4}}$$

$\therefore$ m $(\angle \theta) \simeq 16^\circ\ 19\grave{}$

, $\because$ AB = 5 length unit

$\therefore$ The perpendicular distance = AB $\sin\theta = 5 \sin 16^\circ\ 19\grave{} \simeq 1.4$ length unit.

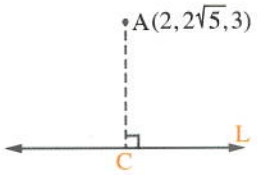
Third solution

Let C be the perpendicular projection drawn from A to the straight line

i.e. C is the projection of the point A on the straight line L

$\therefore C = (-2 + 8t , 3\sqrt{5} - \sqrt{5}t , 1 + 2t)$

$\therefore \overrightarrow{AC} = (-4 + 8t , \sqrt{5} - \sqrt{5}t , -2 + 2t)$ (1)



, $\because \overrightarrow{AC} \perp L$ $\therefore \overrightarrow{AC} \cdot \vec{d} = 0$

$\therefore (-4 + 8t , \sqrt{5} - \sqrt{5}t , -2 + 2t) \cdot (8 , -\sqrt{5} , 2) = 0$

$\therefore -32 + 64t - 5 + 5t - 4 + 4k = 0$ $\therefore 73t = 41 , t = \frac{41}{73}$

$\therefore$ The perpendicular length $= \| \overrightarrow{AC} \|$

$$= \sqrt{\left(-4 + 8 \times \frac{41}{73}\right)^2 + \left(\sqrt{5} - \sqrt{5} \times \frac{41}{73}\right)^2 + \left(-2 + 2 \times \frac{41}{73}\right)^2}$$

$$= \frac{12}{\sqrt{73}} \text{ length unit.}$$

Fourth solution

$\because \overrightarrow{BA} = (4 , -\sqrt{5} , 2)$

$$\therefore \overrightarrow{BA} \times \vec{d} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 4 & -\sqrt{5} & 2 \\ 8 & -\sqrt{5} & 2 \end{vmatrix} = 8\hat{j} + 4\sqrt{5}\hat{k}$$

$$\therefore \text{The perpendicular length} = \frac{\| \overrightarrow{BA} \times \vec{d} \|}{\| \vec{d} \|} = \frac{\sqrt{(8)^2 + \left(4\sqrt{5}\right)^2}}{\sqrt{(8)^2 + \left(-\sqrt{5}\right)^2 + (2)^2}} = \frac{12}{\sqrt{73}} \simeq 1.4 \text{ length unit.}$$